

Artificial Intelligence (AI-2002)

Lecture 4: The Nature of Environments

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The Nature of Environments

- Task environments are the "problems" that rational agents aim to solve.
- Rational agents are "solutions" to these task environments.
- To build rational agents, it's important to first define and understand task environments.
- There are various types of task environments.
- The nature of the task environment influences the design of the agent design.

Specifying the Task Environment

- The rationality of an agent is evaluated based on its performance measure, environment, actuators, and sensors.
- These components are collectively known as the task environment.
- The PEAS description stands for Performance, Environment, Actuators, and Sensors.
- When designing an agent, the first step is to fully specify the task environment.

Let us consider a more complex problem: an automated taxi driver

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

Figure 2.4 PEAS description of the task environment for an automated taxi driver.

Using the PEAS framework, define the task environment for a **Smart Light System** that automatically turns the light on or off based on room occupancy and ambient light conditions. Provide the following details:

- Performance Measures
- Environment
- Actuators
- Sensors

Component	Smart Light System
Performance Measure	Energy efficiency, accurate lighting control, user satisfaction (light turns on/off appropriately)
Environment	Room, kitchen, office
Actuators	Light bulb, Dimmer
Sensors	Motion sensors, Light sensors

Properties of Task Environments

- The range of task environments in AI is vast, but they can be categorized along a small number of dimensions.
- These dimensions significantly influence the appropriate agent design and the applicability of different agent implementation techniques.

Fully observable vs. partially observable:

- If an agent's sensors provide access to the complete state of the environment at every point in time, the environment is fully observable.
- A task environment is partially observable if sensors provide incomplete or inaccurate information, or if some aspects of the environment are missing.
- If an agent has no sensors at all, the environment is unobservable. Though it might seem hopeless, the agent's goals can still sometimes be achieved, even with uncertainty.

Single-agent vs. Multiagent:

- **Single-agent:** An environment where only one agent is involved, such as a crossword puzzle solver.
- **Multiagent:** An environment involving two or more agents, like chess, where the behavior of other agents influences the agent's performance.
- An entity is treated as an agent if its behavior maximizes a performance measure that depends on the agent's actions. For example, in chess, the opponent (entity B) is an agent because its behavior directly impacts the agent's (A's) performance.

Single-agent vs. Multiagent:

- **Competitive:** In a competitive environment like chess, agents aim to maximize their performance while minimizing others.
- **Cooperative:** In a cooperative environment like taxi driving, agents work together (e.g., avoiding collisions), though some competition can still exist (e.g., for parking spaces).

Deterministic vs. Nondeterministic:

- **Deterministic:** The next state of the environment is fully determined by the current state and the agent's action, so there is no uncertainty.
- **Nondeterministic:** If the next state is not fully determined by the current state and action, the environment is nondeterministic.

Deterministic vs. Nondeterministic:

- **Taxi Driving:** A nondeterministic environment, because traffic behavior, tire blowouts, and unexpected events cannot be predicted exactly.
- **Vacuum World:** Often deterministic, but can include nondeterministic elements such as randomly appearing dirt or a malfunctioning suction mechanism.

Stochastic vs. Nondeterministic:

- **Stochastic:** Refers to environments that explicitly model probabilities (e.g., a 25% chance of rain).
- **Nondeterministic:** Refers to environments where possible outcomes are listed, but probabilities are not quantified (e.g., a chance of rain without a specific percentage).

Episodic vs. Sequential:

- **Episodic:** The agent's experience is divided into separate, independent episodes. Each episode involves the agent receiving a percept and performing an action, with no dependencies on previous actions. Example: A part inspection task where each part is considered independently, and the outcome of one episode does not affect the next.
- **Sequential:** The current decision affects future decisions, meaning that actions have long-term consequences. Example: In chess or taxi driving, each move or action can influence the course of the game or journey in the future.

Static vs. Dynamic:

- **Static:** The environment remains unchanged while the agent is deliberating, meaning the agent doesn't need to worry about time passing or changes occurring while it makes decisions.
- **Dynamic:** The environment can change while the agent is deliberating, meaning the agent must constantly respond to changes in the world. If the agent hasn't made a decision, doing nothing still counts as a decision.

Discrete vs. Continuous:

- **Discrete:** The states, time, percepts, and actions can only take on specific, distinct values. For example, chess has a finite number of states, and actions (like moving a piece) are discrete.
- **Continuous:** The states, time, percepts, and actions can take on a continuous range of values. For example, taxi driving involves continuous states (speed, location) and continuous time, where these values change smoothly over time. Actions (like steering) are also continuous.

Known vs. Unknown:

- **Known:** In a known environment, the agent (or designer) knows all the outcomes (or outcome probabilities if the environment is nondeterministic) for every action taken by the agent.
- **Unknown:** In an unknown environment, the agent must learn how the environment works in order to make effective decisions.

Known vs. Unknown:

- The known/unknown distinction is not the same as the fully/partially observable distinction.
- A known environment can still be partially observable, like in solitaire card games where the rules are known, but the cards that are still face down are hidden.
- An unknown environment can be fully observable, like in a new video game where the screen shows the whole game state, but the agent (player) doesn't yet know what the buttons do.

Identify the type of environment it operates in Ludo game.

- **Fully Observable or Partially Observable?**
- **Deterministic or Stochastic?**
- **Single-agent or Multi-agent?**
- **Static or Dynamic?**
- **Episodic or Sequential?**
- **Discrete or Continuous?**

Identify the type of environment it operates in Tic-Tac-Toe Game.

- **Fully Observable or Partially Observable?**
- **Deterministic or Stochastic?**
- **Single-agent or Multi-agent?**
- **Static or Dynamic?**
- **Episodic or Sequential?**
- **Discrete or Continuous?**

Identify the type of environment it operates in Smart Light System.

- **Fully Observable or Partially Observable?**
- **Deterministic or Stochastic?**
- **Single-agent or Multi-agent?**
- **Static or Dynamic?**
- **Episodic or Sequential?**
- **Discrete or Continuous?**

Classify a given task environment

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle Chess with a clock						
Poker Backgammon						
Taxi driving Medical diagnosis						
Image analysis Part-picking robot						
Refinery controller English tutor						

Figure 2.6 Examples of task environments and their characteristics.

Classify a given task environment

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Figure 2.6 Examples of task environments and their characteristics.